NeuralPlane: Multi-view 3D Plane Reconstruction via Neural Fields

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01 NeuralPlane: Authors and Affiliations

- Authors include **Hanqiao Ye**, **Yuzhou Liu**, **Yangdong Liu**, and **Shuhan Shen**.
- Affiliations span the **University of Chinese Academy of Sciences** and the **Institute of Automation, CAS**.
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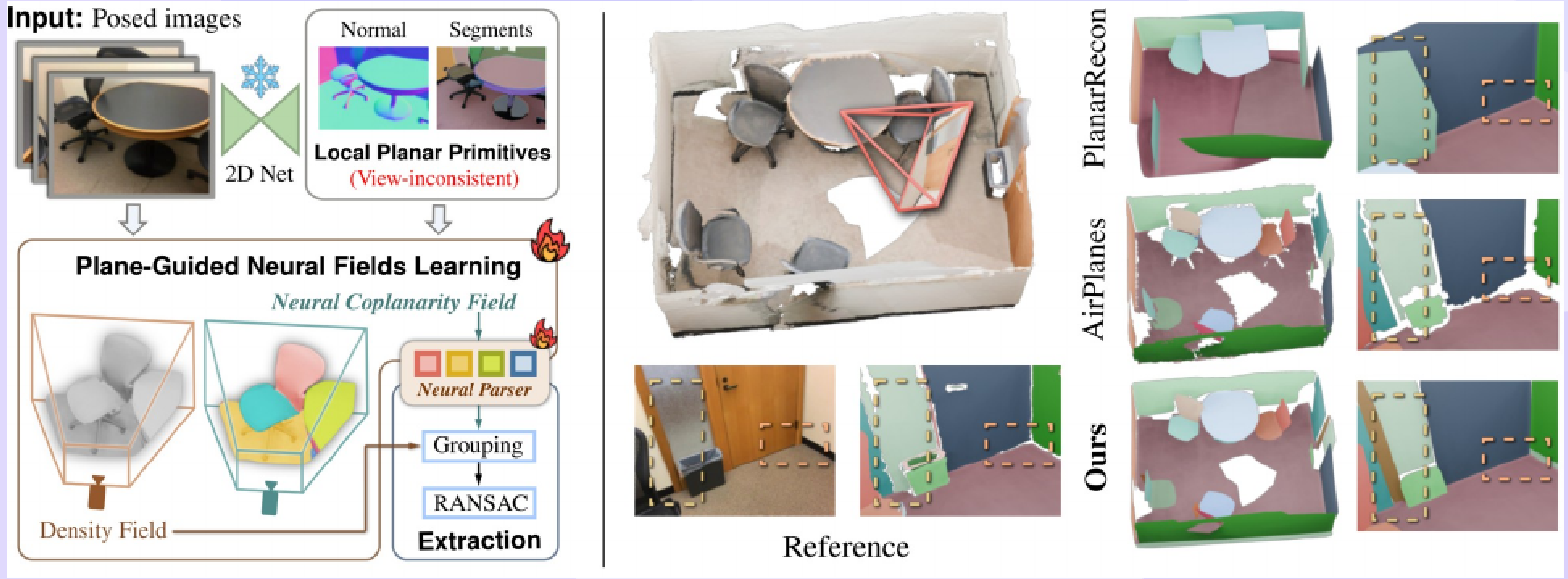
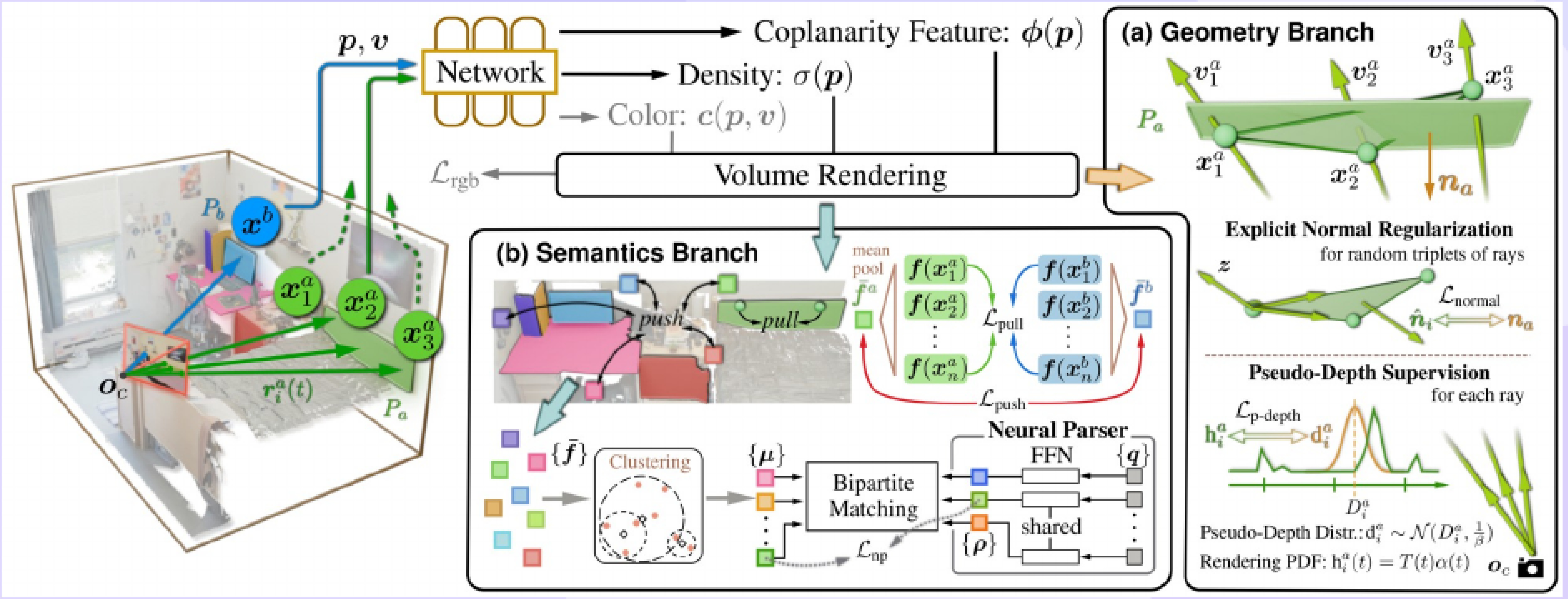
02 NeuralPlane: Multi-view **3**D Plane Reconstruction via Neural Fields



- Teaser shows crisp planes with coherent multi-view semantics.
  - Crisp, semantically aligned multi-view 3D planes
- Method fuses inconsistent 2D planes into a unified 3D coplanar map.
  - Fuses inconsistent 2D plane observations into a unified 3D coplanar map
- Representation enables **compact maps** and expressive structure.

03 Visual Overview: From **2**D Plane Segments to Global **3**D Planes



- Stage 1 extracts monocular plane masks and coarse parameters as local primitives.
  - Monocular self-prompting → local planar primitives
- Stage 2 refines neural geometry with plane constraints and learns coplanarity features.
  - Plane-guided neural optimization + coplanarity features
- Stage 3 groups prototypes, parses instances, and fits planes to produce a mesh.
  - Prototype grouping, instance parsing, plane fitting → mesh

04 Motivation, Problem Setup, and Contributions

- Conventional pipelines decouple geometry and semantics and struggle with noisy inputs.
  - Conventional plane fitting decouples geometry and semantics; struggles with noise
- Goal is compact planar reconstruction of indoor scenes without plane annotations.
  - Structured indoor scenes as compact planar primitives without plane annotations
- Key components include **monocular plane segments**, **plane-guided training**, and **Neural Coplanarity Field**.
  - Monocular plane segments
  - Plane-guided training
  - Neural Coplanarity Field

05 Local Planar Priors: Monocular Segments and Initialization

- 2D plane segments arise from normal clustering refined by SAM (Segment Anything Model) and size filtering.
- Initialization sets mean normal on the mask and estimates offset using SfM (Structure from Motion) keypoints with reliability weights.
- These noisy priors provide useful guidance for joint learning.

06 Plane-Guided Geometry: Normal + Pseudo-Depth

- Normal regularization: fit a local plane to rendered points and align with initialized normals.
- Pseudo-depth: match plane-derived depth to the rendering probability density function (PDF) via Kullback–Leibler (KL) divergence to refine (n, o).
- Joint constraints improve plane placement and **boundary sharpness**.

07 Coplanarity Semantics: Neural Coplanarity Field (NCF), Prototype Parsing, and Global Plane Extraction

- Neural Coplanarity Field learns contrastive features that capture inter-primitive coplanarity.
  - Contrastive features for coplanarity (Lpull/Lpush)
- Neural Parser uses learnable prototypes matched to feature centroids to separate instances.
  - Prototype queries matched to feature centroids
- Extraction groups by prototypes, fits planes with RANSAC, and triangulates a planar mesh.
  - Group by prototypes → RANSAC plane fit → mesh

08 Related Work (brief)

- 3D plane reconstruction often remains local and faces cross-view inconsistency.
  - Local multi-image and stereo methods; limited scene-level fusion
- Video and neural pipelines explore planar primitives and incremental fusion.
  - PlanarRecon; neural fields with planar primitives
- Recent two-stage approaches lift 2D priors to 3D using geometry with robust fitting.
  - AirPlanes lifts 2D priors to 3D via geometry+RANSAC

[Xie'22][Watson'24][Stier'23][Yu'22][Wang'22][Wu'23][Sayed'22][Guo'22][Schonberger'16]

09 Baselines Compared (with source tags)

- PlanarRecon (Xie et al., 2022)
- AirPlanes (Watson et al., 2024)
- FineRecon (Stier et al., 2023)
- MonoSDF (Yu et al., 2022)
- NeuRIS (Wang et al., 2022)
- ObjectSDF++ (Wu et al., 2023)
- SimpleRecon (Sayed et al., 2022)
- ManhattanSDF (Guo et al., 2022)
- COLMAP (Schönberger & Frahm, 2016)
- Full citations in references slide.

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10 Experimental Setup: Datasets, Metrics, Implementation

- Datasets include eight ScanNetv2 scenes and four ScanNet++ scenes.
  - 8 ScanNetv2 + 4 ScanNet++
- Metrics cover geometry accuracy and segmentation consistency with standard thresholds.
  - Geom: Accuracy/Completeness/Chamfer/Precision/Recall (PR) and F1; Seg: Rand Index (RI), Variation of Information (VOI), Segmentation Covering (SC)
  - Segmentation: Rand Index (RI), Variation of Information (VOI), Segmentation Covering (SC)
- Implementation uses Nerfstudio with fast training on a single RTX (NVIDIA RTX GPU) 3090.
  - Nerfstudio; 4k iters; RTX (NVIDIA RTX GPU) 3090

11 Results on ScanNetv**2**: Overall Performance and Insights

- NeuralPlane leads on geometry and segmentation with balanced precision and recall.
  - NeuralPlane leads on geometry and segmentation
  - Balanced precision/recall vs AirPlanes pipelines
- AirPlanes-based pipelines favor missing planes while keeping high precision.
- Qualitative results show **clean planes** and coherent labels across views.

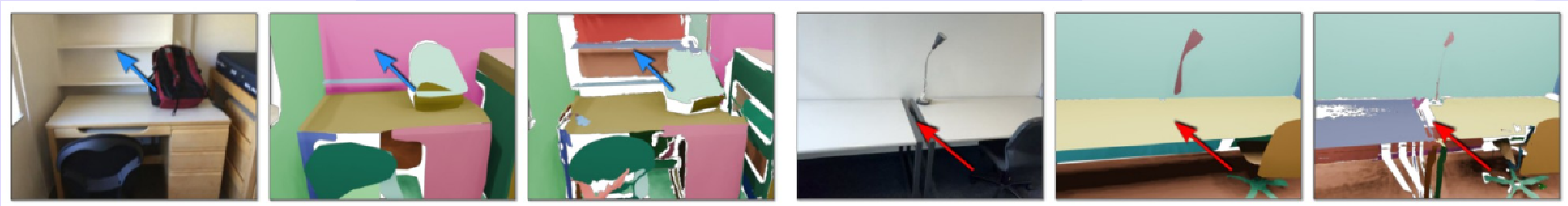
ScanNet++ Generalization

- NeuralPlane achieves competitive geometry and strong segmentation on ScanNet++.
  - Competitive geometry; strong segmentation
- Results remain robust despite sparse keypoints and low texture.
- Comparisons highlight **generalization** without plane annotations.

Efficiency

- Volume-density geometry provides about forty times faster training than implicit SDFs.
  - Volume-density geometry enables ~40× speedup vs implicit SDFs (App. A.4)
- Efficient optimization supports rapid **scene turnaround**.
- Compute efficiency reduces prolonged runtimes in large-scale evaluations.

Main Failure Mode: Noisy **2**D Cues + SfM Bias



- Main failure arises from noisy 2D cues or SfM errors causing missing planes and under-segmentation.
  - Noisy 2D cues/SfM → missing planes, under-segmentation
- Curved or subtly warped surfaces are forced into planar fits and lose detail.
  - Curved surfaces approximated by planes
- Thin or adjacent regions can be over-segmented or merged in challenging settings.
  - Over/under-segmentation on thin/adjacent regions
  - Scope: indoor planar scenes

Ablation Studies: Component Contributions and Prototypes

- DBSCAN (Density-Based Spatial Clustering of Applications with Noise) epsilon strongly influences grouping and **instance count**.
- Small epsilon risks fragmentation of large planes.
- Moderate epsilon improves plane continuity without merging.

Ablating Number of Semantic Prototypes

- Performance remains stable across prototype counts with good detail at thirty-two.
  - Performance stable; Np=32 good detail
- Higher prototype counts can introduce over-segmentation in some scenes.
- Prototype choice controls **level of detail** in planar maps.

Ablation Studies: Component Contributions and Prototypes

- NCF feature dimensionality affects **semantic separability** and training stability.
  - Impact of feature dimensionality
- Low dimensions risk feature collapse and weak boundaries.
- Moderate dimensions yield clean boundaries with compact descriptors.

Ablation Studies: Component Contributions and Prototypes

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  - DBSCAN epsilon affects performance
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Conclusion: NeuralPlane—Joint Plane Geometry and Semantics

- Annotation-free 2D plane cues guide **joint learning** of geometry and semantics.
- Neural Coplanarity Field and parser align clean geometry with instances.
- The method yields compact, high-quality planar maps suitable for **AR and robotics**.

20

References for Baselines

- Xie, Z., et al. (2022). PlanarRecon.
- Watson, J., et al. (2024). AirPlanes.
- Stier, N., et al. (2023). FineRecon.
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Thank You

Questions & Discussion